Crop Bot Regional Yield Screening - Case Study

Contribution

Samuel Faulkner built and checked an interactive corn-hybrid screening application as a graduate hackathon project with team Crop Circuits. The work connected data reconciliation, interactive decision support and supplied-model inference replay.

Recorded work

The application exposes regional predictions, screening thresholds, review flags and downloadable candidate rows. It uses supplied model checkpoints rather than training a new model inside the application.

Recorded outcomes

Reconciled region labels and checked live inference against supplied predictions. Reproduction of supplied predictions is distinct from predictive accuracy against future yields.

Evaluation discipline

The workflow makes retrospective evaluation, baseline behavior and regional limitations visible to reviewers rather than presenting a single headline accuracy claim.

Source and scope

Prepared from the curated PPP Crop Bot Regional Yield Screening experience record. This is a graduate hackathon prototype with historical validation evidence. No model tests were rerun for this portfolio update. Protected competition data and model checkpoints remain private.